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APPLIED RESEARCH

Using Grid-Forming Energy Storage Systems to Provide Dynamic Active Power Support for Hyperscale Data Center

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ABSTRACT Large artificial intelligence (AI) training data centers are emerging as significant, highly dynamic digital loads that pose numerous challenges to power system stability. Methods to study and assess their impact are still being developed and have been the focus of numerous industry interest groups. Intended to contribute to bridging this gap, this paper develops a credible impact study for data centers, presenting a dynamic data-center model that includes an uninterruptible power supply (UPS), co-generator, cooling system, and a grid-forming (GFM) controlled energy storage system (ESS), with explicit converter and control dynamics. Using electromagnetic transient simulations, two stress scenarios are studied: low-voltage faults and fast load fluctuations characteristic of AI training. The results show that an adequately tuned and controlled GFM ESS can significantly improve the large load response by reducing frequency overshoot, raising the nadir, shortening settling time, and enhancing post-fault recovery. Under fast fluctuation AI training load profiles, the ESS also damps active-power oscillations and reduces torsional shaft-torque oscillations in the generator train to safe levels. These findings support ESS-based control as an effective tool for integrating large AI data centers without compromising grid stability.

INDEX TERMS Data center power, energy storage, grid forming, power smoothing, power system faults.

I. INTRODUCTION

To accommodate society's rapidly increasing broadband usage and facilitate development of larger artificial intelligence (AI) models, particularly large language models, hyperscale data centers with consumptions exceeding 1 GW are rapidly connecting to the grid [1], [2]. Being predominantly electronic loads, hyperscale data centers are both sensitive to grid disturbances and capable of rapidly varying their power consumption, especially when they are used for large-scale parallel computing tasks (e.g., training large language models). Due to the data centers' size and tendency to concentrate geographically, these characteristics lead to rapid variations in active power flows

in the transmission system. These variations can exceed the response capabilities of traditional technologies such as electromechanical generation, leading to adverse impacts to transmission system reliability [3]. Energy storage systems (ESS), due to their power electronic grid interface and flexible dispatch, are well suited to provide dynamic active and reactive power support necessary to maintain a reliable grid while integrating hyperscale data centers [4]. Grid-forming (GFM) control is especially applicable to ESS because the DC link inherently supplies the necessary energy buffer, enabling GFM functionality to be realized primarily through software [5].

A. LITERATURE REVIEW AND MOTIVATION

While hyperscale data centers with power consumptions exceeding 1 GW are rapidly being connected to bulk

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power systems, accurate dynamic models that simultaneously capture the interactions among information technology equipment (ITE), cooling systems, uninterruptible power supplies (UPS), ESS, and co-located generation remain scarce. Existing studies typically represent data centers as aggregate static or quasi-static loads [6], [7], [8], which is useful for flexible load management but does not resolve fast electromechanical and control driven dynamics. Other works examine specific aspects of data center power systems—for example, energy consumption and power electronics challenges from building distribution through DC architectures down to the board level [9], an all-in-one load data center power emulator on a reconfigurable power-electronics hardware testbed [10], or an isolated parallel-ring bus datacenter microgrid with multiple UPSs [11]. However, a high-fidelity dynamic model that integrates ITE, cooling, UPS, ESS, and co-located generation in a system-level context is still lacking. To address this gap, we develop such an integrated dynamic model in this work.

As AI workloads scale, hyperscale data centers are introducing new operational challenges that can delay or constrain their interconnection to the bulk power system. In 2024, a 230 kV transmission line fault in the Eastern Interconnection led to a customer-initiated, simultaneous loss of approximately 1,500 MW of voltage sensitive load that operators had not anticipated [12], [13]. Many large loads include protection and control functions that disconnect during disturbances; the resulting tripping and reconnection can cause significant frequency excursions and increase the risk of generator tripping, thereby threatening overall grid stability and integrity. In addition, rapid load ramps (up or down) of data center loads can stress the system because generation must be quickly loaded or unloaded in response. Large loads can also act as sources of oscillations, introducing reliability risks [14]. Against this backdrop, there is growing industry interest in using GFM ESS as a mitigation measure for AI-driven data center load variability and associated power oscillations. For example, EPC Power is positioning GFM ESS solutions for load variability smoothing and for reducing the risk of load shedding during grid disturbances [15]. Tesla has similarly highlighted the use of grid-forming ESS for AI training load smoothing and improved low-voltage ride-through capability [16].

Motivated by this research gap and real-world applications, our work focuses on how a GFM ESS can be used to mitigate the adverse dynamic impacts of large AI data centers. The main objectives are to: (i) develop a high-fidelity dynamic model of a hyperscale AI data center and its interaction with the bulk grid; (ii) quantify the impacts of large load tripping and reconnection events on system frequency and voltage; and (iii) assess the effect of fast AI-driven load variability on torque fatigue in the co-located generator at the data center.

B. CONTRIBUTIONS

In this work, case studies are performed using electromagnetic transient (EMT) analysis with a topologically small

but detailed transmission system equivalent and a detailed data center model. Two specific use cases are considered: post-fault frequency regulation and oscillation suppression. For post-fault frequency regulation, the transmission system equivalent provides an aggregate representation of the transmission system's fault and frequency response. For oscillation suppression, a steam turbine with a detailed multi-mass turbine model is placed at the data center for the purposes of generation. While modeled in the same facility, our findings also apply to any synchronous generator that could be nearby, e.g., thermal base generation. This enables analysis of the internal stresses placed on the turbine shaft, which are one of the primary reliability concerns associated with rapidly varying data center loads. For both cases, we demonstrate GFM ESS control strategies are capable of mitigating the reliability issues and characterize the effects of major system and design variables such as system inertia and ESS size. These results illustrate the potential efficacy of ESS for augmenting bulk system reliability and provide insights into the credible application and performance expectations for ESS.

The contributions of this work are:

- development of a high-fidelity dynamic model of a hyperscale data center, integrating ITE, cooling systems, UPS, ESS, and a co-located generator within a PSCAD framework, capturing their control dynamics and interactions;
- system-level assessment of large data center integration, including the impacts of large load tripping and re-connection on bulk system frequency and voltage, and evaluation of a grid-forming ESS providing dynamic active and reactive power support following faults, demonstrating improved post-fault frequency nadir/overshoot and voltage regulation;
- analysis of high-frequency load-induced oscillations, showing how fast AI/data-center-type load fluctuations can stress the co-located generator (e.g., increased torque oscillations), and evaluation of the effectiveness of the ESS in suppressing these oscillations.

The remainder of this paper is organized as follows. Section II describes the test system model. Section III presents case studies in which the ESS provides dynamic support following a fault and under fast frequency fluctuations caused by AI training loads. Section IV concludes the paper.

II. DESCRIPTION OF THE SYSTEM UNDER STUDY

In this section, we present the test system model, which consists of: a synchronous machine (SM) connected to the transmission grid, a hyperscale data center with a centralized UPS, a synchronous generation plant collocated with the data center, and an ESS located at the data center.

Fig. 1 illustrates the test system used in this study. The system equivalent is modeled using a detailed synchronous generator equipped with a GOV2 governor [17], a steam turbine, and an AC1A exciter model [17]. Impedances

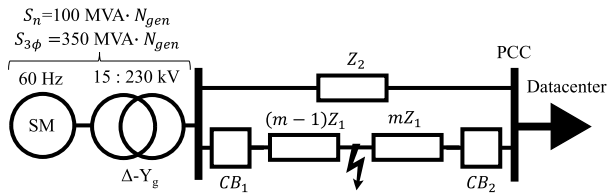


FIGURE 1. Single-line diagram of transmission system model.

Z_1 and Z_2 represent two transmission system paths to the data center. For a given study, the magnitudes of Z_1 and Z_2 are selected to obtain a desired short-circuit ratio (SCR), both with CB_1 and CB_2 closed (SCR) and with them open (SCR').

A. HYPERSCALE DATACENTER MODEL

In this work, we consider a data center architecture with a centralized UPS, which is one of the most common configurations in large facilities. The overall data center model is intentionally generic (i.e., not vendor-specific) and was developed based on feedback from a broad group of industry experts, whose input helped ensure that the architecture, load composition, and control assumptions are consistent with current industry practice. The model we developed in this work is not intended to be an exact representation of any specific data center; rather, it is designed as a reasonable and transparent starting point for the development of site-specific models suitable for grid-level studies.

Fig. 2 illustrates the component level models within the data center. This arrangement is intended to be an aggregate representation of a cluster of large data centers built to provide concurrent maintainability. Single-phase and three-phase rectifiers represent to represent representative motor and auxiliary loads on the cooling and auxiliary systems, and ITE is connected to the grid via a double-conversion UPS.

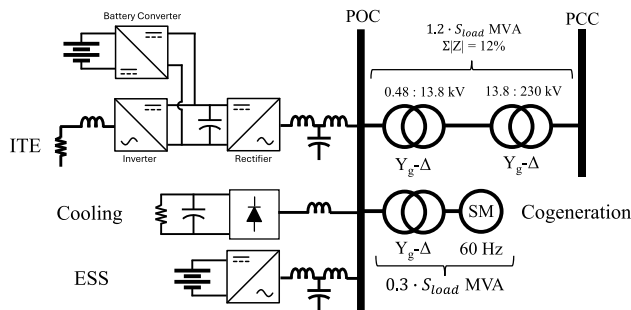


FIGURE 2. Single-line diagram of datacenter model.

Fig. 3 illustrates the control diagram for the UPS power converters. Both the UPS rectifier and inverter are voltage-source converters represented by an average value models (AVM). The input rectifier control is grid-following (GFL) with current vector control; the active current refer-

ence is controlled by a proportional–integral (PI) based DC bus voltage regulator, and the reactive current reference is controlled by a reactive power regulator with a power reference of 0 MVar, giving unity power factor operation. Rectifier current is limited to 1.25 pu using a dq -domain current limiter with active power prioritization. The output inverter control is GFM with droop control; the active power reference is set equal to 0, and the reactive power reference is controlled by a PI based AC voltage regulator. The inverter current magnitude is limited to 1.5 pu using hysteresis-based current clipping [18].

The UPS battery is represented as a DC voltage source with a magnitude of 800 V and an internal resistance of 5 mΩ. The battery converter is a bidirectional buck–boost converter. The modulation index of the converter is controlled using both an active power regulator (to manage charging and discharging) and a DC voltage regulator. Coordination between the two DC link voltage regulators is achieved by operating the battery converter at a slightly lower DC voltage reference (0.99 pu) and limiting the DC voltage regulator to contribute only to the boost operating mode. With this arrangement, the DC voltage regulator of the battery converter participates actively in DC voltage regulation when the DC link voltage is below 0.99 pu. This allows the UPS to maintain tight DC voltage regulation even when the rectifier is operating at its current limits, thereby minimizing the effect of deep voltage sags in the rectifier AC voltage on the inverter AC voltage.

B. COGENERATION (SYNCHRONOUS GENERATION) MODEL

A synchronous generation (SG) plant is collocated with the data center to illustrate both cases of a cogeneration plant or a nearby base generator, such as a thermal plant. The cogeneration plant is modeled as a large high-speed turbine, such as might be used for nuclear or natural gas generation. The inclusion of grid-connected generation with hyperscale data centers is becoming increasingly common as it can reduce the need for transmission system upgrades which would otherwise significantly extend the interconnection timeline (e.g., from 2 years to 5 to 10 years). Additionally, generation-rich areas are naturally attractive options for the connection of hyperscale data centers, meaning that a hyperscaler is likely to be located close to turbine generation even if the data center itself does not host any co-generation. This case study serves as a reasonable approximation of both cases.

A six-mass turbine model is used, providing a representation of the major oscillatory modes within the turbine shaft. Turbine parameters are from a real steam turbine [19]. The turbine governor and exciter control systems are not modeled; while they are indeed of import in practical studies, as they can either improve or worsen torsional interactions [20]. Designing and tuning control systems for the considered turbine is beyond the scope of the study.

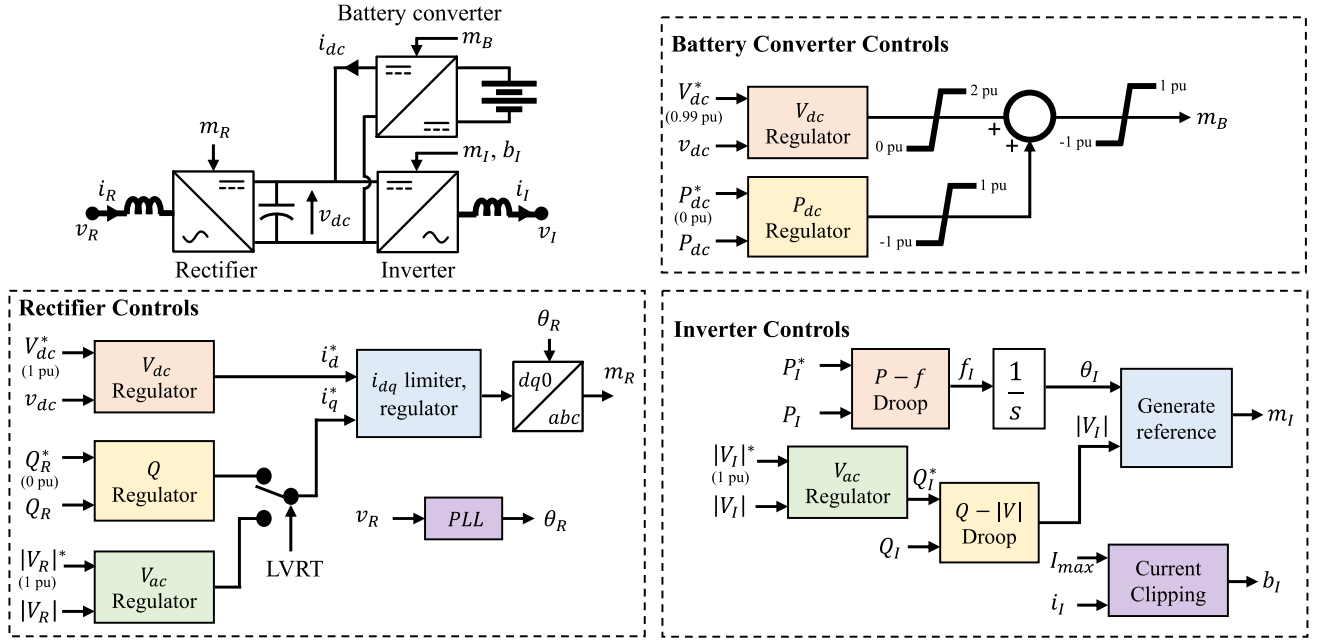


FIGURE 3. Uninterruptible power supply control overview.

The aging of the steam turbine in response to cyclical loads, such as those imposed by AI training loads, may be quantified using an stress/life-cycle (SN) Curve, which provides the number of cycles that a given component can withstand before experiencing mechanical failure. In turbine generators, this failure is usually manifest as a crack in the turbine shaft and effectively marks the end of the turbine's useful life. Fig. 4 illustrates the exemplar SN curve used to perform fatigue assessments in this study; this is intended to be a reasonable generic SN curve for a steam turbine—specific SN curves differ depending on the turbine and shaft section in question and are protected proprietary information held by turbine manufacturers. When designing a solution to suppress

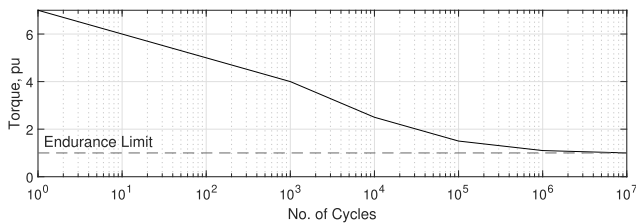


FIGURE 4. SN curve used to evaluate turbine shaft fatigue from cyclical AI training load.

oscillations that might cause torsional interactions, the goal is to ensure that the torques within the generators of concern have amplitudes below the black dashed line in Fig. 4, which indicates the high cycle fatigue limit (HFCL). Torque ripple below this threshold has no measurable effect on turbine shaft life.

C. BATTERY ENERGY STORAGE SYSTEM MODEL

An aggregated battery ESS is located with the data center. For the ESS inverter, we employ a virtual synchronous machine (VSM)-based grid-forming control. The implementation corresponds to the WECC-approved generic REGFM_B1 model, which was jointly developed by Pacific Northwest National Laboratory (PNNL) and several GFM control manufacturers. This model is designed to capture the characteristic dynamic behavior of VSM-type GFM ESS installations. The detailed model specification and parameter ranges are documented in [21]. The VSM controller incorporates the power-frequency droop characteristic as well as inertia and damping response functions. The control block diagram is shown in Fig. 5. The inverter angle expressed as $\frac{d\delta_{\text{vsm}}}{dt} = \omega_{\text{vsm}}$, and the internal angular frequency dynamics are given by,

$$2H_{\text{vsm}} \frac{d\omega_{\text{vsm}}}{dt} = P_{\text{ref}} - P - \left(\frac{1}{m_p} + D_1 + \frac{sD_2}{s + \omega_D} \right) (\omega_{\text{vsm}} - \omega_0), \quad (1)$$

where H_{vsm} , m_p , D_1 , D_2 , ω_D , ω_{vsm} , ω_0 , P_{ref} , P denote inverter inertia constant, $P - \omega$ droop coefficient, damping constant, transient damping constant, angular frequency of the washout filter, angular frequency, nominal angular frequency, active power reference, and measured active power.

The terminal voltage of the ESS inverter E is captured by

$$k_{iv} \frac{dE}{dt} = V_{\text{ref}} + m_q(Q_{\text{ref}} - Q), \quad (2)$$

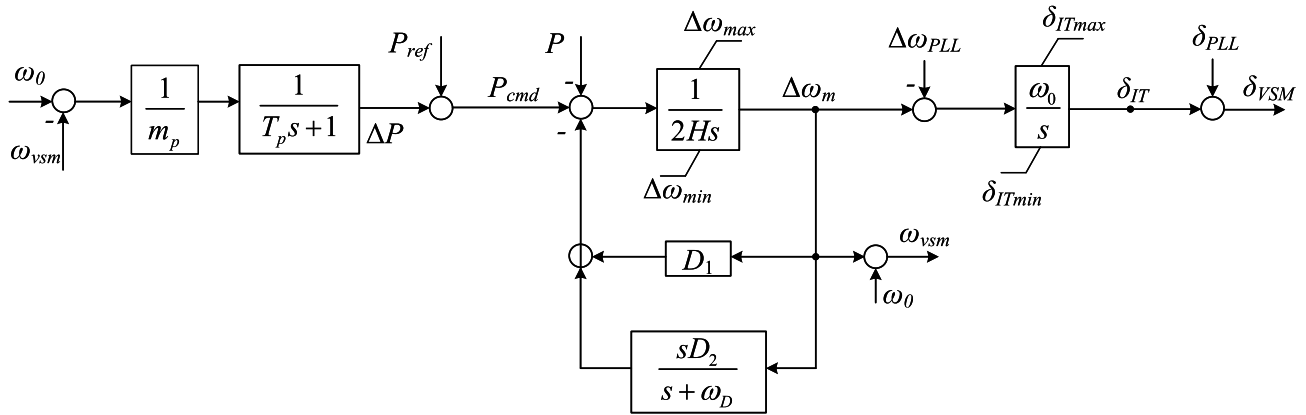


FIGURE 5. Phase angle generation of the virtual synchronous machine control [21].

where k_{iv} , V_{ref} , m_q , Q_{ref} , Q denote the integral gain, voltage reference, $Q - V$ droop coefficient, reactive power reference, and measured reactive power.

To limit steady-state active and reactive power, the voltage magnitude variation is constrained by the steady-state reactive current limit, and the phase-angle variation is constrained by the steady-state active current limit. A selectable PQ priority algorithm (P- or Q-priority) is used to allocate the steady state current between active and reactive components. The PWM layer applies current clipping to cap transient over-current during faults. Additionally, to ensure the inverter remains synchronized after fault clearing, a phase-locked loop (PLL) is used to lock the inverter internal phase angle to prevent loss of synchronism. This is achieved by feeding forward the grid voltage angle measured by the PLL (δ_{PLL}) as shown in Fig. 5.

III. CASE STUDIES

In this section, two main applications of dynamic support for data centers are considered: fast frequency response and oscillation suppression. In both scenarios, the key capability provided by the ESS is fast response and the ability to supply significant active power.

A. POST-FAULT FREQUENCY REGULATION

In this case study, we evaluate the ability of the ESS to stabilize the bulk system frequency following the tripping of the data center load in response to a transmission system fault. This is a similar scenario to the gigawatt-scale load drops experienced in both the Texas and Eastern Interconnections [12], [13]. In the following studies, SCR is 6 and SCR' is 3. The synchronous generators in Fig. 1 have a total nominal power of 8 GW with a 5% governor droop. A 2.5 GW constant-power load is connected on the SG side. The data center includes 140 MW of cooling load and 1.1 GW of ITE load. Assume a three-phase voltage fault occurs on the transmission line and lasts 320 ms. Most transmission faults are cleared in less than 100 ms and so

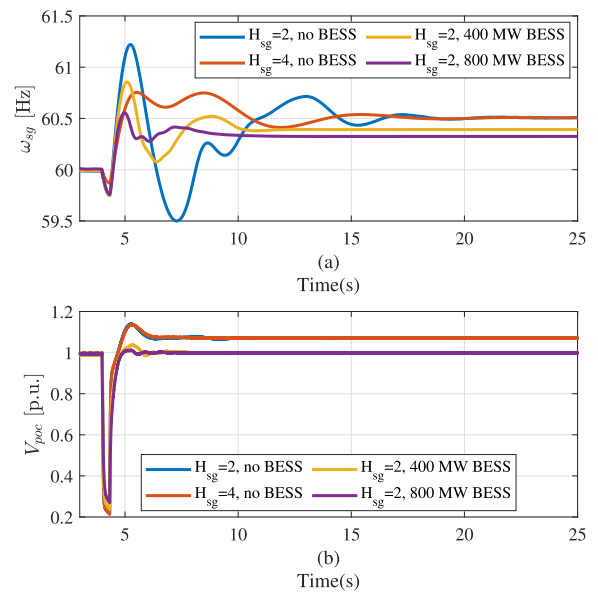


FIGURE 6. Effects of SG inertia on, (a) frequency, (b) POC voltage, in response to fault-induced data center tripping.

this case represents delayed clearing as might be caused by a failed circuit breaker or communications link. The rectifiers for the cooling loads and the UPS use undervoltage relays that disconnect if the POC voltage remains below 0.8 p.u. for more than 50 ms. We consider scenarios with and without automatic reconnection of the UPS rectifier; when enabled, it recloses once the POC voltage has recovered to 1.0 p.u. for at least 1 s.

1) WITHOUT AUTOMATIC UPS RE-CONNECTION AFTER FAULT

The system inertia constant is varied to emulate grids with different inertia levels. When an ESS is included, its settings are: $m_p = 0.01$, $H_{vsm} = 10$, $D_1 = 20$, $D_2 = 200$, $\omega_D = 50$. Figure 6 shows the post-fault frequency and the data center

POC voltage across inertia levels, with and without a ESS. Lower inertia yields larger frequency excursions and slower recovery: the frequency peak reaches 61.22 Hz at $H_{sg} = 2$ s, compared with 60.75 Hz at $H_{sg} = 4$ s. An inertia value of $H_{sg} = 4$ s approximates a system dominated by traditional synchronous generation. A value of $H_{sg} = 2$ s approximates a system with roughly 50% penetration of inverter-based resources, with the assumption that the assumption that these resources do not participate in inertial response. In the low-inertia case $H_{sg} = 2$ s, adding an ESS reduces the frequency peak to 60.86 Hz with a 400 MW rating and to 60.55 Hz with an 800 MW rating and shortens the settling time. In the no-ESS case, the post-fault data-center POC voltage settles at 1.07 pu rather than 1.0 pu because a 350 MVA shunt capacitor is installed at the POC for steady-state voltage support. When the data center load trips after the fault, the capacitor overcompensates and the bus voltage rises to 1.07 p.u. With a ESS, the capacitor is unnecessary; the ESS provides reactive power support, maintaining both pre- and post-fault voltages at 1.0 p.u.

We study a low-reserve scenario with 4 GW online generation (inertia $H = 4$ s). A 1.5 GW load is connected upstream, and the data center parameters are unchanged. Fig. 7 illustrates the effect of ESS capacity on grid frequency rise following fault-induced tripping of the data center load. Figure 7 shows the grid-frequency rise after fault-induced tripping of the data-center load for different ESS power ratings. Without ESS, the frequency peaks at 61.98 Hz and settles at 61.00 Hz. Adding ESS reduces the overfrequency: with a 400 MW ESS, the peak drops to 61.62 Hz and the steady-state frequency to 60.73 Hz; with an 800 MW ESS, the peak further drops to 61.25 Hz and the steady-state frequency to 60.46 Hz. Fig. 7 illustrates how the overfrequency conditions compare to the over frequency ride-through (FRT) limits defined in NERC PRC-024 [22]. The limits vary by interconnection; values for the Western Interconnection (WI) are chosen here simply for illustration. Observe that, without a ESS (0 MW of capacity), the overfrequency condition exceeds the limits that generators are required to ride through. This indicates a serious reliability risk—such a condition could cause widespread tripping of generation units and subsequent cascading outages. A 400 MW ESS is just sufficient to bring the excursion within the tolerable limits, while an 800 MW ESS provides a more robust margin, including against potential future reductions in system inertia and unforeseen contingencies.

Under the same base case with a 4 GW synchronous machine, a 1.5 GW constant-power load, and an 800 MW ESS, we adjust the ESS control parameters to study their impact on the frequency response to fault-induced data center tripping. Fig. 8 shows the results of varying the ESS inertia constant H with $m_p = 1\%$, $\omega_D = 50$. It can be observed that changing the inertia constant does not significantly affect the frequency behavior. The secondary frequency rise is slightly improved when the inertia constant is 10 compared to when

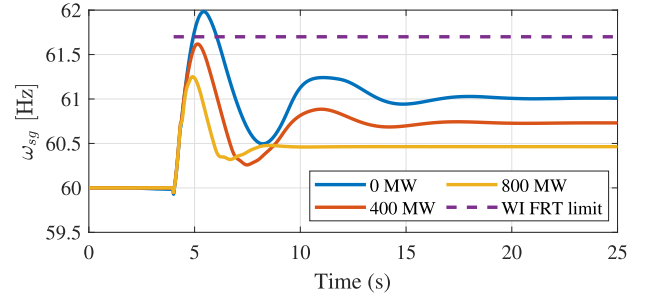


FIGURE 7. Relation of frequency rise from fault-induced tripping of data center load to NERC PRC-024 overfrequency ride-through requirements.

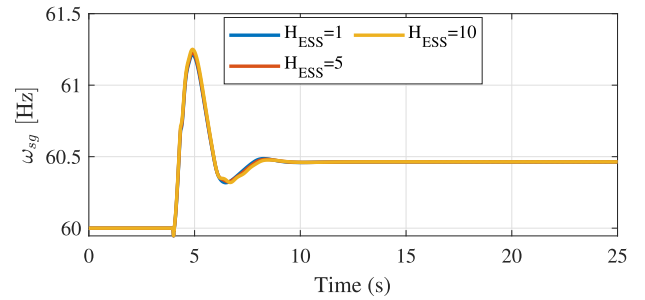


FIGURE 8. Effects of ESS inertia on frequency in response to fault-induced data center tripping.

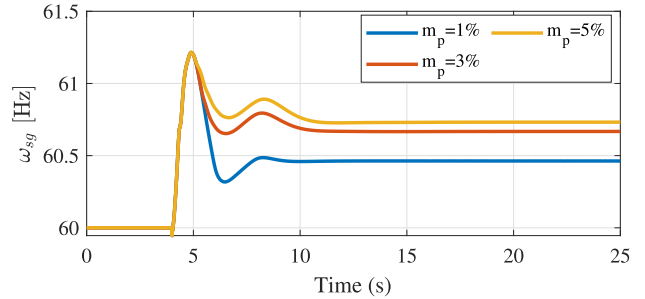


FIGURE 9. Effects of ESS droop coefficient on frequency in response to fault-induced data center tripping.

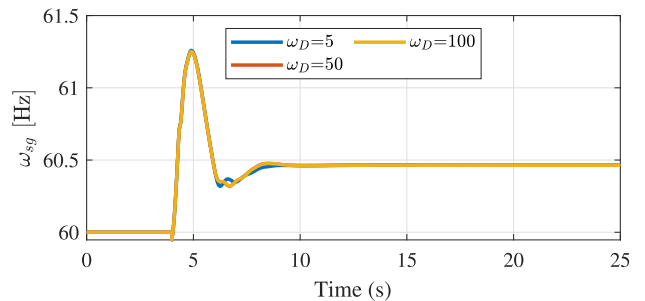


FIGURE 10. Effects of ESS ω_D on frequency in response to fault-induced data center tripping.

it is 1 or 5. With $H = 1$, $\omega_D = 50$, we vary the active power-frequency droop coefficient of the ESS m_p , and the corresponding frequency response is shown in Fig. 9. It can be observed that changing the m_p significantly affects the

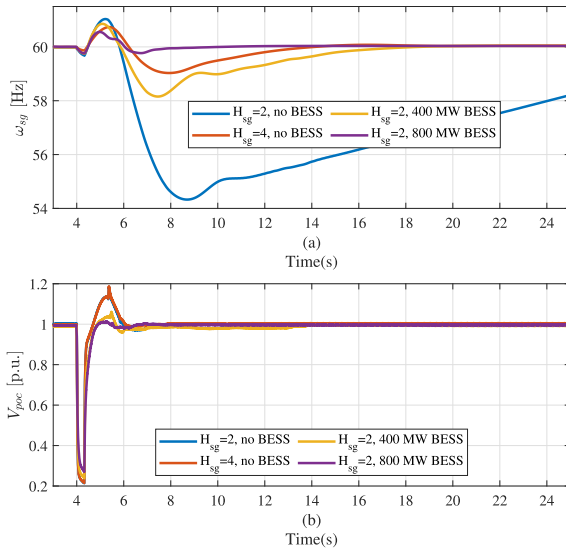


FIGURE 11. Effect of SG inertia on, (a) frequency, (b) POC voltage.

frequency recovery: the steady state frequency is 60.73 Hz for $m_p = 5\%$, 60.67 Hz for $m_p = 3\%$, 60.46 Hz for $m_p = 1\%$. In addition, the smaller the m_p is, the shorter the frequency recovery time becomes. With $m_p = 1\%$, $H = 10$, we vary the angular frequency on the washout block ω_D , and the corresponding frequency response is shown in Fig. 9. It can be observed that changing ω_D does not significantly affect the frequency behavior. The frequency recovery is slightly better damped for larger values of ω_D .

2) WITH AUTOMATIC UPS RE-CONNECTION AFTER FAULT

With the UPS re-connection function enabled, Fig. 11 shows the post-fault SG frequency and the data-center POC voltage across inertia levels, with and without an ESS. Without an ESS, the data center load drop following the fault produces a frequency overshoot: the peak reaches 61.033 Hz at $H_{sg} = 2$ s and 60.737 Hz at $H_{sg} = 4$ s. In the low-inertia case ($H_{sg} = 2$ s), a 400 MW ESS reduces the peak to 60.554 Hz, and an 800 MW ESS further reduces it to 60.500 Hz, demonstrating fast frequency response and the value of ESS in low inertia systems. Reconnecting the UPS after the fault produces a frequency dip: the frequency nadir is 54.33 Hz at $H_{sg} = 2$ and the frequency nadir is 59.03 Hz at $H_{sg} = 4$. This suggests that, in a low inertia system, the simple approach of quickly reconnecting the load post fault may be insufficient to avoid significant frequency fluctuations. The low-inertia case $H_{sg} = 2$ violates the frequency capability curve specified by NERC PRC-024. A ESS addresses this issue by raising the post-fault frequency nadir to 58.16 Hz with a 400 MW rating and to 59.77 Hz with an 800 MW rating. Meanwhile, the settling time is reduced with the ESS allocation.

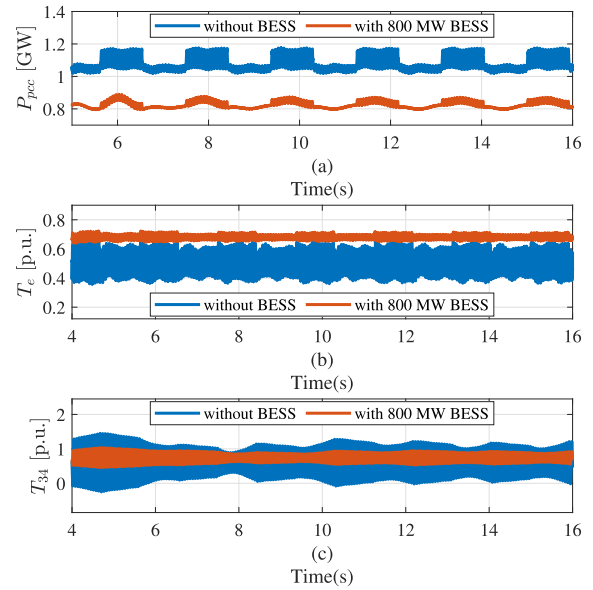


FIGURE 12. Response to fast AI-training load fluctuations (16 Hz), (a) data-center PCC active power, (b) co-generator electrical torque, (c) torsional shaft torque (masses 3–4).

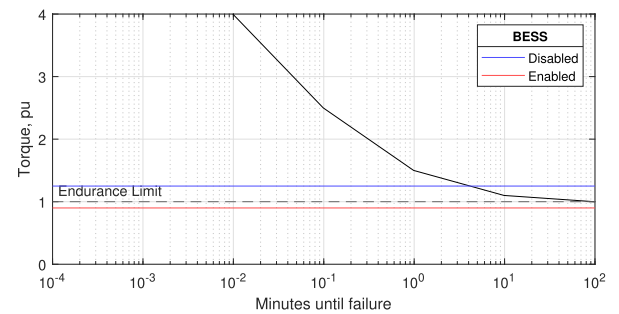


FIGURE 13. Effects of ESS on turbine aging introduced by 90 to 100% AI training load with fast frequency of 16 Hz.

B. OSCILLATION SUPPRESSION FOR COGENERATION OPERATION

Available AI training load data indicate that typical hyperscale AI workloads exhibit two dominant time scales: (i) a “slow” component on the order of seconds, corresponding to aggregate job/mini-batch dynamics, with characteristic frequencies roughly in the 0.1–1 Hz range, and (ii) a “fast” component on the order of milliseconds, associated with accelerator/cluster-level activity, with dominant components broadly in the 5–30 Hz range. In this application, we evaluate the effects of rapid, periodic variations in data center active power consumption on the co-generator. The concern is that these variations will cause torques in the turbine that exceed its mechanical limits, resulting in severe damage to a generation asset that is critical to transmission system reliability. Consideration is given to recent advances in ITE-level controls, such as software-level mitigations and

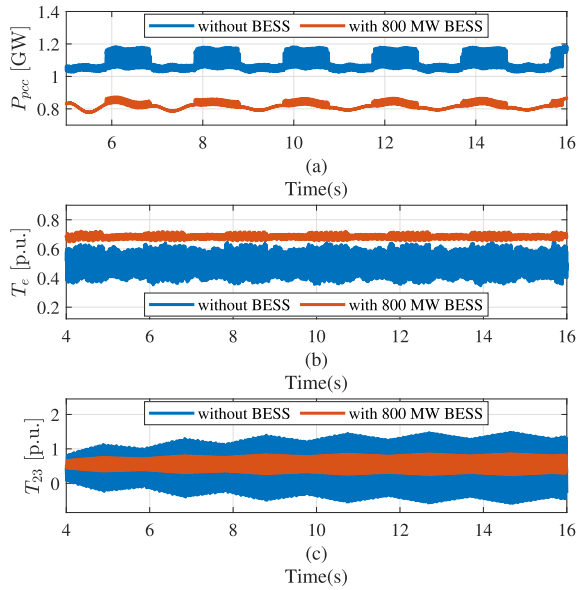


FIGURE 14. Response to fast AI-training load fluctuations (25.6 Hz), (a) data-center PCC active power, (b) co-generator electrical torque, (c) torsional shaft torque (masses 2–3).

graphics processing unit (GPU) smoothing, which can somewhat reduce the amplitude of these fluctuations [23]. Our results indicate that, even with these improvements, additional oscillation suppression, such as that provided by the ESS, is necessary to avoid turbine damage.

The co-generator is rated at 372 MVA. The AI training load has a nominal power of 1.1 GW, with a fast component at 16 Hz and a slow component at 0.5333 Hz; the resting load is 0.99 GW. A frequency of 16 Hz is selected to provide conservative results as it coincides with a lightly damped shaft mode. The slow component is selected to be roughly an order of magnitude lower and an integer divisor of the fast frequency. Fig. 12 shows the data-center PCC active power, the co-generator electrical torque, and the torsional shaft torque between masses 3 and 4. Although other shaft sections also experience significant torque, the 3–4 section is the most severe; for brevity, only this torque is plotted. Physically, it corresponds to the torque between the two low-pressure turbine stages. Fig. 12 shows that adding an 800 MW ESS at the data center significantly reduces oscillations in PCC active power, co-generator electrical torque, and the torsional torque between masses 3 and 4.

The torque waveforms illustrated in Fig. 12 can be used to perform an initial assessment of the turbine's ability to withstand the torque oscillations without experiencing shaft damage. That said, SN curves are intended to capture the fatigue induced by a simple periodic load and do not fully capture how the turbine will wear when responding to waveforms with broadband subsynchronous frequency content such as AI training waveforms. Thus, we must make some conservative assumptions:

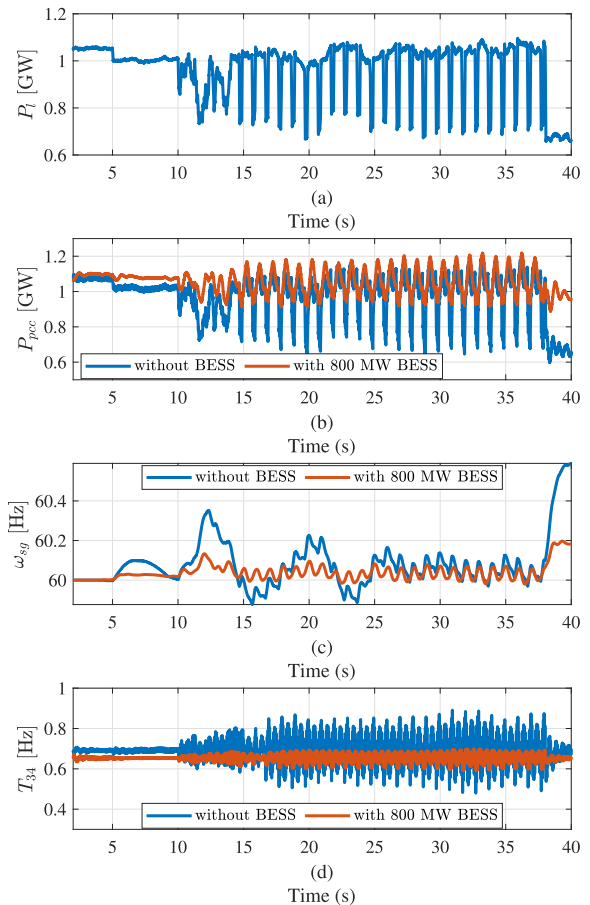


FIGURE 15. Response to an example AI data center load profile from [3], (a) AI training load profile, (b) active power at PCC, (c) frequency, (d) torsional shaft torque (masses 3-4).

- Each cycle of the fast frequency in the AI training load is treated as one cycle in terms of the SN curve.
- The peak steady state torque ripple is the value evaluated against the SN curve
- The SN curve is derated linearly based on the average torque loading. In other words, if the average torque on the generator is 0.5 pu, the SN curve introduced in Fig. 4 is shifted downwards by 0.5 pu

Employing these assumptions, Fig. 13 illustrates the effects of the torques imposed by various AI training loads on the turbine life, both with and without the ESS included. Observe that the reduction in torque provided by the ESS is sufficient to achieve a value below the endurance limit, minimizing the risk that the AI training load profile will meaningfully affect turbine life. Additionally, observe that, with the relatively high frequency of the AI loading, the region where the torque ripple ages the shaft but does not result in a near-term failure is small (torque ripples here are all near 1 p.u.). The short time to failure associated with torque ripples in the range of 1 to 1.5 pu suggests that all that is needed to determine whether an oscillation is acceptable is

the HFCL threshold—loadings that are even slightly over the HFCL translate to turbine lifetimes of mere minutes or hours. This contrasts with some historical sources of turbine fatigue, such as line switching [24].

Under an IT load profile with fast (25.6 Hz, the frequency of another lightly damped shaft mode) and slow (0.5115 Hz) variation components, Fig. 14 shows the simulated data center PCC active power, cogenerator electrical torque, and the torsional response of shaft sections II–III, with and without ESS deployment. The results indicate that the BESS mitigates oscillations under these fast frequency fluctuations.

In addition to the periodic variations in data center active power consumption, we also verify the ESS's effectiveness in smoothing an approximation of a real AI data center load profile [3]. Figure 15(a) shows the AI training load profile, which has been linearly scaled to a peak value of 1.1 GW. Note that, in practice, fluctuations from individual feeds will exhibit some level of destructive interference, making this a conservative test. Using this profile, we simulate the following quantities at the data center PCC and the co-located generator: (b) active power at the PCC, (c) system frequency, and (d) torsional shaft torque (between masses 3–4 of the generator shaft) both without ESS and with an 800 MW ESS providing grid-forming support. The results in Figs. 15(b–d) demonstrate that: Without ESS, the PCC experiences rapid, irregular power fluctuations driven by the AI training load, which translate into noticeable frequency variations and elevated torsional torque oscillations. With the ESS in service, the fast power fluctuations at the data center PCC are substantially mitigated. As a result, the frequency variations induced by the AI training load are reduced, and the torsional torque oscillations in the co-generator are also significantly damped.

IV. CONCLUSION

An EMT data-center model—including a UPS, cogenerator, cooling system, and ESS—that explicitly represents power converters and their control dynamics was developed in this work. Using this model, we assess the impact of data center low-voltage ride-through and examine how fast-varying data-center AI training loads stress the cogenerator. Under low-voltage fault conditions, EMT simulations show that protection-driven tripping of data-center loads can cause frequency excursions that violate the NERC PRC-024 frequency capability curve, risking generator trips. Deploying an ESS substantially mitigates these impacts—raising the frequency peak/nadir and shortening the settling time—thereby improving post-fault frequency recovery and preventing potential widespread outages. The ESS also suppresses high-frequency AI-training load fluctuations, which in turn reduces torsional shaft-torque oscillations in nearby generators to below the HFCL and avoids significant reductions in turbine shaft life.

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